Google Image Scraping and Data Collection – Problem Statement

You are tasked with developing a Python-based solution to automate the process of collecting images from Google Images based on a specific query. The objective is to extract, process, and store relevant image data efficiently for further analysis or machine learning use cases. Manual image collection is time-consuming and inefficient, especially when large datasets are required. Therefore, an automated approach is essential.

Objective

1. Automate image collection from Google Images
2. Extract and store image data based on search queries
3. Organize images into structured directories
4. Enable scalable dataset creation for ML and data analysis

Key Tasks / Questions

1. Send HTTP requests to retrieve image data
2. Parse HTML content to extract image tags
3. Download images efficiently using extracted URLs
4. Store images in a structured folder system
5. Automate scraping for multiple queries
6. Handle errors such as blocked requests
7. Optimize performance and efficiency

Challenges

1. Handling request blocking using User-Agent
2. Parsing dynamic HTML structures
3. Managing duplicate or irrelevant images
4. Ensuring reliable downloads

Expected Outcome

1. Automated Python script for image extraction
2. Organized image dataset stored locally
3. Efficient data collection compared to manual methods